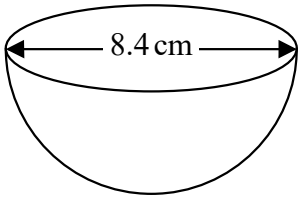
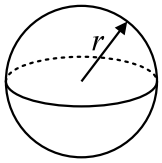
19 The diagram shows a hemisphere with diameter 8.4 cm.



Volume of sphere = $\frac{4}{3}\pi r^3$



Work out the volume of the hemisphere.
Give your answer correct to 3 significant figures.

..... cm^3

(Total for Question 19 is 2 marks)

20 $d = \frac{1}{8}c^3$

$c = 10.9$ correct to 3 significant figures.

By considering bounds, work out the value of d to a suitable degree of accuracy.
Give a reason for your answer.

10.85 ≤ c < 10.95 from 3 significant figures.

Lower d = $10.85^3/8 \approx 159.6611$.

Upper d = $10.95^3/8 \approx 164.1165$.

Both bounds round to 160 to nearest ten.

So d = 160 to a suitable degree of accuracy.

4 marks total
Bounds for c
Cube both / divide by 8
d bounds 159.66–164.12
Both give 160 to nearest 10

(Total for Question 20 is 4 marks)

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